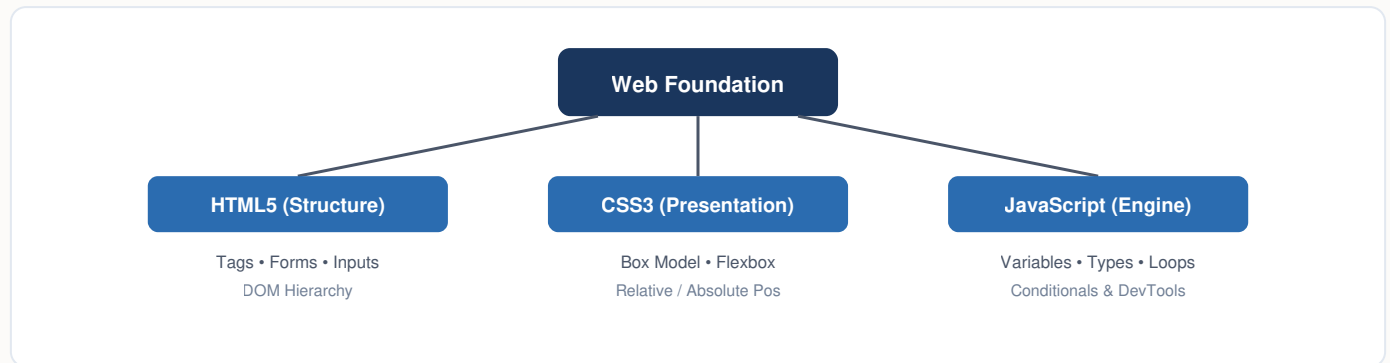


Web Foundations: HTML, CSS & JavaScript

Sheryians Full-Stack Crash Course Notes • Quick Revision & Mindmaps for ML/Full-Stack Development

1. Architectural Architecture & Mindmap

Web applications in full stack ML systems operate as three tightly coupled layers: semantic structure, layout presentation, and logical runtime.



2. HTML5: Semantic Structure & Form Capturing

HTML provides the skeleton. In ML dashboards and AI interfaces, forms and input tags are critical because they collect features prior to model dispatch.

Core Structural Elements

- `<!DOCTYPE html>` : Defines standard HTML5 document mode.
- `<h1>` to `<h6>` : Structural headings hierarchy.
- `<p>` , `<span>` , `<div>` : Paragraphs, inline segments, and block divisions.
- `<ul>` , `<ol>` , `<li>` : Bulleted and ordered listings.
- `` : Media embedding with descriptive fallback text.

Data Collection Tags (ML Focus)

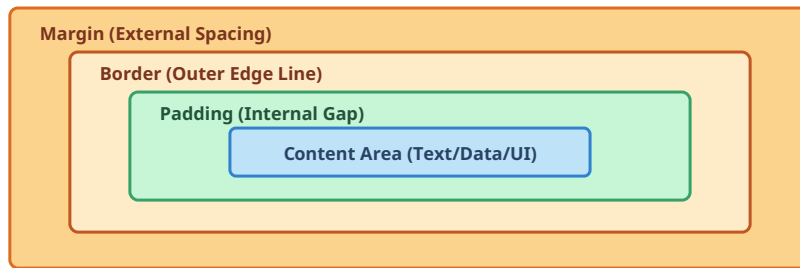
- `<form>` : Container wrapper for submission events.
- `<input type="text">` : String inputs (e.g. prompt, job role).
- `<input type="number">` : Numeric parameters (e.g. salary, experience, age).
- `<button type="button">` : Trigger client-side action without page reload.

```
<!-- Standard Prediction Input Pattern -->
<div class="input-card">
  <label for="experience">Years of Experience:</label>
  <input type="number" id="experience" placeholder="e.g. 3" />
  <button id="predict-btn">Predict Salary</button>
</div>
```

3. CSS3: Box Model, Positioning & Flexbox Layouts

The CSS Box Model

Every element in a webpage is rendered as a concentric rectangular box consisting of four distinct segments:



Golden Rule: Reset & Box Sizing

Always include `* { box-sizing: border-box; margin: 0; padding: 0; }` at the top of your stylesheet so margins and paddings do not break dimension calculations.

Positioning Attributes

Property	Behavior
<code>static</code>	Normal page document flow (default).
<code>relative</code>	Offset relative to self; anchor for absolute children.
<code>absolute</code>	Removed from document flow; positioned relative to nearest relative parent.
<code>fixed</code>	Pinned to viewport; remains fixed during scroll.
<code>z-index</code>	Stacking context integer (higher renders on top).

Flexbox Essentials

Property	Usage
<code>display: flex</code>	Enables flexible layout for direct children.
<code>flex-direction</code>	<code>row</code> (horizontal) or <code>column</code> (vertical).
<code>justify-content</code>	Aligns along main axis (<code>center</code> , <code>space-between</code>).
<code>align-items</code>	Aligns along cross axis (<code>center</code> , <code>flex-start</code>).
<code>gap</code>	Even gutter spacing between flex items.

4. JavaScript: Execution, Types & Logic

Variable Declarations

- `const` : Immutable binding. Preferred default declaration for references and constants.
- `let` : Block-scoped mutable variable. Used in loops and updating state accumulators.
- `var` : Legacy function-scoped variable. Avoid in modern codebases.

Primitive Data Types

- `string` : `"FastAPI"` , `'Classification'`
- `number` : Integers & floats (e.g. `42` , `0.98`)
- `boolean` : `true` or `false`
- `undefined` : Variable declared but uninitialized
- `null` : Explicit intentional empty value

Developer Tools & Logging

```
// Logging & Monitoring
console.log("Feature vector:", [2.5, 4]);
console.warn("Latency high on model call");
console.error("Endpoint 500: Server Error");
```

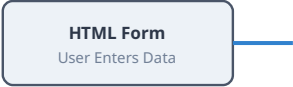
Control Flow (Branching)

```
const score = 82;
if (score >= 90) {
  status = "High Match";
} else if (score >= 70) {
  status = "Moderate Match";
} else {
  status = "Low Match";
}
```

5. End-to-End Client Execution Flow

Visualizing how user actions trigger evaluation logic before sending data to an ML backend:

HTML Form
User Enters Data



Hands-on Checkpoint Project: Candidate Screening Portal

Objective: Build a simple centered card using `display: flex`. Collect two inputs (`Experience` and `Technical Score`). On button click, evaluate if `exp > 2 && score >= 75`. Render dynamic feedback text directly on the screen.